

## Experiment No.: 1

### Experiment Title: Basics of Microcontroller

Group No.: B6

Name: Pritam Priyadarshi

Roll no.: 230793

#### Results for the group task:

| Task no. | Topic             | Task                                                                                                                                                                                                                                               | Report                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Introduction      | Get to know these pins: IOREF, RESET, GND, 5V, 3.3V, Vin, A0-A5, Aref, Rx0-Tx1, PWM pins, and Built-in LED pin.                                                                                                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2        | Digital Read      | Use the provided program for Task 2 to digital-read the pin 8 and print it on the serial monitor. Connect the following pins to pin 8 and print the result on the serial monitor: none (open), GND, RESET, IOREF, AREF, 3.3V, 5V                   | (a) The following are the values when pin 8 is connected to: <ul style="list-style-type: none"><li>• Open (none) → 0</li><li>• GND → 0</li><li>• RESET → 0</li><li>• IOREF → 1</li><li>• AREF → 0</li><li>• 3.3V → 1</li><li>• 5V → 1</li></ul> (b) Behavior: <ul style="list-style-type: none"><li>i). Connected to GND then left unconnected → Always showed 0.</li><li>ii). Connected to 5V then left unconnected → First showed 1 then 0.</li></ul> |
| 3        | Digital Read      | Use the provided program for Task 3 to turn on the built-in LED only if pin 8 is LOW (connected to GND) and turn off otherwise. (in the code, the command <code>pinMode(8, INPUT_PULLUP)</code> maintains the pin 8 at HIGH when left unconnected) | <ul style="list-style-type: none"><li>• LED turned ON when logic LOW is applied to pin 8.</li><li>• LED turned OFF when the logic LOW is removed from pin 8 and remained unconnected.</li></ul>                                                                                                                                                                                                                                                         |
| 4        | Digital Read      | Introduce a five-second delay in the loop of the last task.                                                                                                                                                                                        | NO, the LED did not turn ON immediately when pin 8 is given a LOW signal, because <code>delay(5000)</code> causes it to wait for 5ms.                                                                                                                                                                                                                                                                                                                   |
| 5        | Digital Read      | Write a code to alternatively switch pin13 (LED) to HIGH and LOW with a delay of 1 second in between.                                                                                                                                              | Voltage reading of pin 13 using multimeter when: <ul style="list-style-type: none"><li>• LED is on: 5.09V</li><li>• LED is off: 0V</li></ul>                                                                                                                                                                                                                                                                                                            |
| 6        | Analog Write(PWM) | Write a code to apply a PWM signal on pin 9. Read the voltage at pin 9 using a multimeter. Tweak the argument inside                                                                                                                               | Observations: <ul style="list-style-type: none"><li>• <code>analogWrite(255)</code>: 5.06V</li><li>• <code>analogWrite(127)</code>: 2.52V</li></ul>                                                                                                                                                                                                                                                                                                     |



- **GND:**

[illegible]

- **RESET:**

[illegible]

- **IOREF:**

[illegible]

- **AREF:**

[illegible]

- **3.3V & 5V:**

[illegible]